

An Analytic Hindered Rotor Model for Calculating Microcanonical Variational Unimolecular Rate Constants from Reaction Path Properties

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Abstract

A model is proposed for performing microcanonical variational transition state theory calculations which incorporates ideas from vibrator and flexible variational transition state models. Vibrational frequencies, moments of inertia, and potential energy for the variational transition state are found by reaction path following as for the vibrator model. However, the transitional modes are treated as hindered rotors using an analytic potential and an analytic density of states, which are fit to barriers for hindered rotation determined from reaction path following. The model proposed here differs from the flexible transition state model in that the density of states for the transitional modes is analytic and transitional modes and external rotational angular momenta are uncoupled. For the $\text{H} + \text{CH}_3 \rightleftharpoons \text{CH}_4$ system, rate constants calculated with this new model are only 6–23% smaller on average from those of the flexible transition state model for values of total angular momentum which correspond to average rotational temperatures of 0–2000 K. Harmonic frequencies calculated for the transitional modes from the hindered rotor Hamiltonian are in good agreement with the exact values found by a reaction path analysis. © 1994 John Wiley & Sons, Inc.

Introduction

Unimolecular bond fission reactions, without a potential energy barrier for the reverse association reaction, are important processes in both combustion and atmospheric kinetics. Variational Rice–Ramsperger–Kassel–Marcus (RRKM) theory has proven to be a very useful method for calculating rate constants as a function of energy and angular momentum for these unimolecular reactions [1–3]. If there is extensive vibration/rotation coupling, only the total angular momentum quantum number J is treated as a constant of the motion [4–13]. For the situation where this coupling is weak and the symmetric top model is adequate, the rotational quantum number K may also become a constant of the motion [7–9].

Two general approaches have been advanced for calculating the variational RRKM rate constant from an analytic potential energy surface [3,8,14]. For the first approach, the rate constant is evaluated from properties of the reaction path [15]. This is the steepest descent path, s , in mass-weighted Cartesian coordinates, and associated with it are principal moments of inertia $I_i(s)$ for the molecular system and $3N-7$ harmonic vibrational frequencies $\nu_i(s)$ for motions orthogonal to the path [16,17]. These $I_i(s)$ and $\nu_i(s)$ are then inserted into the rigid-rotor/harmonic-oscillator formalism to calculate the minimum in the sum of states vs. s and locate the transition state and determine its properties. The resulting transition state is called a vibrator transition state since each of its $3N-7$ degrees of freedom is treated as a vibration [3,8,18–22]. Besides this model's assumption of harmonicity and vibrational/rotational separability, another

possible shortcoming is its treatment of modes that become rotations and translations in the products, i.e., transitional modes [23], as harmonic oscillators. At large internuclear separations the frequencies for these modes approach zero and, thus, their density of states diverges instead of becoming the correct constant value for a collection of free rotors.

In the second approach, which is called flexible transition state theory (FTST) [2,24–26], the transitional modes are assumed to be represented by the Hamiltonian

$$(1) \quad H_{tr} = T_{r1} + T_{r2} + l^2/2\mu R_{cm}^2 + V$$

and their density of states $\rho(E_{tr}, J)$ is calculated using classical mechanics. In eq. (1) the T_r terms are rotational kinetic energies for the two fragments, l is the orbital angular momentum, μ is the fragments' reduced mass, R_{cm} is the fragment-fragment center-of-mass separation, and V , which depends on R_{cm} and the relative orientation of the two fragments, is the intermolecular potential. The unimolecular system's total angular momentum J is the vector sum of the rotational angular momenta j_1 and j_2 for the two fragments and the orbital angular momentum l . The above approach for treating the transitional modes is reminiscent of the orbiting transition state model, widely applied to ion-molecule systems [27].

Flexible transition state theory identifies the remaining modes as conserved modes and treats them as quantum harmonic oscillators, as is done in the vibrator transition state model. Initially, frequencies for the conserved modes were assumed to be the same as those for the separated fragments [2,24–26]. However, in more recent work, frequencies for the conserved modes have been equated to the $\nu_i(s)$ obtained from a reaction path analysis as described above [28,29]. Flexible transition state theory was originally formulated with the assumption that the reaction path is simply the fragment center-of-mass separation R_{cm} [2]. A more general reaction path, based on the Hamiltonian in eq. (1), has been recently formulated [30–32].

Properties of the flexible transition state are determined by finding the minimum in the sum of states along the reaction path. The sum of states is written as the convolution

$$(2) \quad N^\ddagger(E, J) = \int_0^{E^\ddagger} N_v(E^\ddagger - E_{tr}) \rho_{tr}(E_{tr}, J) dE_{tr}$$

where $N_v(E^\ddagger - E_{tr})$ is the quantum mechanical harmonic sum of states for the conserved modes, $\rho_{tr}(E_{tr}, J)$ is the classical density of states for the transitional modes, and $E^\ddagger$ is the total energy E minus the reaction path potential. The density of states for the transitional modes is evaluated by a Monte Carlo integration [2]. In recent work this integration has been greatly simplified [11–13].

Implicit in the transitional mode Hamiltonian, eq. (1), is the assumption of rigid fragments. Given this description, FTST provides an exact classical accounting of the transitional modes, which are represented as anharmonic, hindered rotors. The angular momentum of these rotors is properly coupled with the orbital angular momentum. However, as fragments associate, coupling between the transitional and conserved modes (which is explicitly neglected in FTST) is expected to become increasingly significant. From a normal mode perspective, the transitional modes begin to combine linearly with the fragments vibrational coordinates of proper symmetry to form the incipient molecular normal modes of vibration. Closely associated with such coupling is nonrigidity of the fragments. These considerations raise questions about the relative accuracy of eq. (1) at small separations, but no quantitative assessment has yet been undertaken.

In the work reported here we present a model for calculating variational unimolecular rate constants which consists of elements from both the vibrator and flexible transition state models. A major goal of this work is to remove the divergence in the transitional modes density of states present in the vibrator transition state model, but retain the overall simplicity of this model. A motivating factor for pursuing the model proposed here is the recent work by Jordan, et al. [33] in which analytic densities of state are given for hindered rotors.

Model

For the model proposed here conserved modes are treated as quantum harmonic oscillators, as is done in both the vibrator and flexible transition state models. To illustrate the proposed treatment of transitional modes, consider the two simple bond fission reactions $\text{CH}_4 \rightarrow \text{H} + \text{CH}_3$ and $\text{C}_2\text{H}_6 \rightarrow 2\text{CH}_3$. The first reaction has three transitional modes; a H + CH₃ relative separation and a two-dimensional rotation for the CH₃ moiety. The latter reaction has six transitional modes; a CH₃ + CH₃ relative translation, a H₃C—CH₃ torsion and a two-dimensional rotation for each CH₃ moiety. The model proposed here assumes the relative separation is associated with the reaction coordinate, the torsion is a free-rotor, and the two-dimensional rotation of a fragment moiety is treated classically with the Hamiltonian

$$(3) \quad H = \frac{1}{2I} [p_\theta^2 + p_\phi^2/\sin^2\theta] + V(\theta, \phi)$$

This Hamiltonian has also been used for two-dimensional rotations of transitional modes in previous work [18,20–22,33]. For the calculations reported here the proposed model is applied to $\text{CH}_4 \rightarrow \text{H} + \text{CH}_3$.

The classical density of states for the Hamiltonian in eq. (3) is

$$(4) \quad \rho(E) = \frac{1}{h^2} \int_0^\pi d\theta \int_0^{2\pi} d\phi \int_{-\infty}^\infty dp_\theta \int_{-\infty}^\infty dp_\phi \delta(H-E)$$

The Hamiltonian's partition function is

$$(5) \quad Q(T) = \frac{1}{h^2} \int d\Gamma \exp(-H/kT)$$

where $d\Gamma = d\theta d\phi dp_\theta dp_\phi$. The density of states and partition function are related by the Laplace transform

$$(6) \quad Q(T) = \int_0^\infty dE \rho(E) \exp[-E/kT]$$

For an ion-dipole dissociation $V(\theta, \phi)$ in eq. (3) only depends on θ . Similarly, Pacey and co-workers [34–36] found that an accurate rate constant for $\text{H} + \text{CH}_3 \rightarrow \text{CH}_4$ association is obtained when $V(\theta, \phi)$ is replaced by $V(\theta)$. Based on these results, Jordan et al. [33] considered the following $V(\theta)$ for representing the potential in eq. (3):

- (i) $V(\theta) = 0$ (free rotor)
- (ii) $V(\theta) = (V_o/\pi)\theta$ (linearly hindered rotor)
- (iii) $V(\theta) = 1/2 V_o(1 - \cos \theta)$ (sinusoidally hindered rotor, i.e., ion-dipole potential)
- (iv) $V(\theta) = V_o \sin^2 \theta$ (Pacey's function)
- (v) $V(\theta) = 0$ for $\theta \leq \theta_{hs}$
 $= \infty$ for $\theta > \theta_{hs}$ (free rotor with hard-sphere potential walls)
- (vi) $V(\theta) = (V_o/\pi^2)\theta^2$ (harmonically hindered rotor)

In each case V_o represents the barrier height of the potential function.

Jordan et al. [33] found an analytic partition function and an analytic density of states for each of the above $V(\theta)$ potentials except the $\sin^2\theta$ function used by Pacey and co-workers [34–36]. Though there is an analytic partition function for this latter potential, as pointed out previously [36], it can only be represented as an infinite series. However, as shown in the Appendix, the potential has a simple analytic expression for the density of states which is

$$(7) \quad \begin{aligned} \rho(E) &= 8\pi^2 I / \sigma h^2, & E \geq V_o \\ \rho(E) &= [8\pi^2 I / \sigma h^2] [1 - (1 - E/V_o)^{1/2}], & E < V_o \end{aligned}$$

where σ is the symmetry number for the two-dimensional rotation.

The transition state is found variationally by locating the minimum in the sum of states along the reaction path, which, as described in the Introduction, is the path of steepest descent in mass-weighted cartesian coordinates. The sum of states is determined by convoluting the quantum sum of states for the conserved modes and the classical density of states for the transitional modes as is done in eq. (2) for the FTST model [37]. However, the model described here is different from the FTST model in its treatment of angular momentum. If the K quantum number for a symmetric (or near-symmetric) top is assumed to be a constant of motion for the transition state [9], the sum of states is

$$(8) \quad N^\ddagger(E, J, K) = \int_0^{E^\ddagger} N_v(E^\ddagger - E_{tr}) \rho(E_{tr}) dE_{tr}$$

where $E^\ddagger$ is the internal energy at a position along the reaction path and is given by

$$(9) \quad E^\ddagger = E - E_o - E_r^\ddagger(J, K)$$

Here E_o is the potential energy (including zero point energy) and $E_r^\ddagger(J, K)$ is the external rotational energy for the position on the reaction path. Both E_o and the moments of inertia for calculating $E_r^\ddagger(J, K)$ are found by reaction path following. The molecular system is treated as a symmetric or near-symmetric top for calculating $E_r^\ddagger(J, K)$. Zero point energy along the reaction path is that for the conserved modes, which are treated quantum mechanically.

The sum of states for the conserved modes, $N_v(E^\ddagger - E_{tr})$ in eq. (8), is calculated using the Beyer–Swinehart algorithm [38]. The conserved modes are treated as quantum harmonic oscillators with frequencies determined by reaction path following. The analytic expression used for the classical density of states of the transitional modes, $\rho(E_{tr})$ in eq. (8), depends on the form used for $V(\theta)$. If $V(\theta)$ is a free rotor, a linearly hindered rotor, a sinusoidally hindered rotor, or a free rotor with hard-sphere potential walls, $\rho(E_{tr})$ is taken from the compilation of Jordan et al. [33]. However, for $V(\theta) = V_o \sin^2\theta$, $\rho(E_{tr})$ is given by eq. (7).

Coriolis coupling can mix the $2J + 1$ K levels for a particular value of J and destroy K as a good quantum number [7–9]. Maintaining the rigid rotor model, but allowing mixing of the $2J + 1$ K levels results in $2J + 1$ mixed levels each with the same sum of states denoted by

$$(10) \quad N_K^\ddagger(E, J) = \sum_{K=-J}^J N^\ddagger(E, J, K) / (2J + 1)$$

where the $N^\ddagger(E, J, K)$ are from eq. (8).

Calculation for $\text{CH}_4 \rightleftharpoons \text{H} + \text{CH}_3$

In this work the above model is applied to CH_4 decomposition and its reverse reaction $\text{H} + \text{CH}_3$ association. The analytic potential energy surface [21,39,40] used for these calculations has been tested in previous work. Classical trajectory calculations [40] and calculations with vibrator [21] and flexible [28] variational transition state models for this surface give similar rate constants for $\text{H} + \text{CH}_3 \rightarrow \text{CH}_4$ association, which reproduce the still uncertain experimental rate constants [41–43] to within a factor of two. The microcanonical variational calculations reported here can be compared with previous vibrator and flexible microcanonical variational calculations for the $\text{CH}_4 \rightleftharpoons \text{H} + \text{CH}_3$ system [8].

For this work, harmonic vibrational frequencies, moments of inertia and potential energy are required for the $\text{CH}_4 \rightleftharpoons \text{H} + \text{CH}_3$ system as a function of the reaction path, and this information is known from previous calculations [21,44]. The moment of inertia for the hindered rotor was determined from the reduced moment of inertia expression [45]

$$(11) \quad I = I_{\text{CH}_3}(1 - I_{\text{CH}_3}/I_{\text{H}---\text{CH}_3})$$

where I_{CH_3} is the moment of inertia for the CH_3 rotor and $I_{\text{H}---\text{CH}_3}$ is the moment of inertia for the $\text{H}---\text{CH}_3$ system [46]. The moment of inertia for $\text{H}---\text{CH}_3$ is found from the Cartesian coordinates along the reaction path. CH_3 was treated as a planar ring [29] when calculating I_{CH_3} . Equation (11) was developed by Pitzer and Gwinn to treat internal rotational motions in molecules. However, if the unimolecular fragments are treated as rigid-rotors, it has been shown [29,46] that I_{CH_3} should be used for the hindered rotor moment of inertia instead of I . Equation (11) is used here in an ad hoc manner to account for neglected couplings between the transitional modes of the two fragments and/or between transitional modes and conserved modes (e.g., the actual rotors are nonrigid). As pointed out by Aubanel, et al. [29] additional work clearly needs to be done to test eq. (11). The effect of using I_{CH_3} instead of I in the unimolecular rate constant calculations is considered below.

The potential energy for the transitional modes has been represented as a hindered rotor by Aubanel et al. [29]. They fit $V(\theta)$ for rotation of CH_3 by the Pacey function $V_0 \sin^2\theta$ and determined V_0 relative to the reaction path potential. The resulting V_0 were fit to

$$(12) \quad V_0 = C \exp(-AR^2)$$

where R is the $\text{H}---\text{C}$ distance on the reaction path and $C = 164 \text{ kcal/mol}$ and $A = 0.43 \text{ \AA}^{-2}$. The density of states for the transitional modes $\rho(E_{tr})$ was calculated from eq. (7) using eq. (12) for V_0 . The sum of states for a position along the reaction path, with the assumption of mixed K levels, is then found from eq. (10) [47]. The sum of states for the transition state is the minimum in this sum along the reaction path.

In Table I values for the transition state sum of states determined here at different values of E and J are compared with previously obtained results [8] for the flexible transition state model and a model which treats the transitional modes as harmonic oscillators. Also given are results determined from the formalism described above with $V(\theta) = 0$, so that the transitional modes are free rotors. The sum of states is largest for this model. Table I shows that there are substantial variations in the ratio between the flexible and hindered rotor sums of states vs. E and J . These variations are not understood at this time.

TABLE I. Comparison of $\text{CH}_4 \rightleftharpoons \text{H} + \text{CH}_3$ transition state sum of states for different treatments of the transitional modes.

$E(\infty)^b$	J^c	Sum of States ^a			
		Flexible ^d	Hindered ^e	Harmonic ^f	Free ^g
0.443	0	6.31	6.13 (7.12) ^h	5.50	7.46
	10	1.03×10^2	$1.11 (1.26) \times 10^2$	1.02×10^2	3.38×10^2
	20				
	30				
	40				
0.709	0	12.00	8.48 (10.21)	9.38	11.86
	10	7.81×10^2	$7.52 (8.75) \times 10^2$	6.48×10^2	13.20×10^2
	20				
	30				
	40				
1.181	0	21.21	17.35 (20.25)	15.89	20.67
	10	2.89×10^3	$2.91 (3.42) \times 10^3$	2.62×10^3	3.88×10^3
	20	1.89	2.79 (3.20)		73.3
	30				
	40				
1.784	0	40.39	20.10 (26.24)	26.13	39.09
	10	6.92×10^3	$7.16 (8.47) \times 10^3$	5.96×10^3	8.96×10^3
	20	2.08×10^3	$1.44 (1.71) \times 10^3$	1.21×10^3	4.44×10^3
	30				
	40				
2.974	0	90.20	71.22 (87.09)	64.96	90.61
	10	2.15×10^4	$2.05 (2.42) \times 10^4$	1.74×10^4	2.57×10^4
	20	1.64×10^4	$1.48 (1.77) \times 10^4$	1.22×10^4	2.56×10^4
	30	1.33×10^2	$3.74 (4.42) \times 10^1$		8.25×10^2
	40				
3.568	0	123.40	97.17 (115.82)	91.84	120.71
	10	3.18×10^4	$3.03 (3.70) \times 10^4$	2.51×10^4	3.85×10^4
	20	2.98×10^4	$2.74 (3.33) \times 10^4$	2.23×10^4	4.46×10^4
	30	2.65×10^3	$1.53 (1.84) \times 10^3$	1.64×10^3	8.36×10^3
	40				
5.738	0	3.41×10^2	$2.70 (3.26) \times 10^2$	2.52×10^2	3.35×10^2
	10	9.72×10^4	$8.84 (10.9) \times 10^4$	7.88×10^4	1.10×10^5
	20	1.24×10^5	$1.20 (1.47) \times 10^5$	1.03×10^5	1.75×10^5
	30	4.71×10^4	$4.12 (5.10) \times 10^4$	3.43×10^4	9.22×10^4
	40	3.82×10^2	$1.15 (1.40) \times 10^2$		2.77×10^3
5.947	0	3.73×10^2	$2.99 (3.84) \times 10^2$	2.71×10^2	3.86×10^2
	10	1.08×10^5	$9.42 (11.5) \times 10^4$	8.51×10^4	1.22×10^5
	20	1.41×10^5	$1.37 (1.66) \times 10^5$	1.16×10^5	1.99×10^5
	30	5.62×10^4	$5.14 (6.38) \times 10^4$	4.01×10^4	1.07×10^5
	40	1.06×10^3	$4.16 (5.07) \times 10^2$	5.47×10^2	6.10×10^3
7.434	0	6.36×10^2	$4.80 (5.89) \times 10^2$	4.64×10^2	6.69×10^2
	10	1.92×10^5	$1.70 (2.15) \times 10^5$	1.55×10^5	2.28×10^5
	20	2.90×10^5	$2.91 (3.56) \times 10^5$	2.45×10^5	4.11×10^5
	30	1.48×10^5	$1.36 (1.69) \times 10^5$	1.09×10^5	2.56×10^5
	40	1.82×10^4	$1.15 (1.45) \times 10^4$	9.85×10^3	5.62×10^4
8.921	0	1.03×10^3	$8.19 (10.2) \times 10^2$	7.59×10^2	1.01×10^3
	10	3.24×10^5	$2.96 (3.53) \times 10^5$	2.57×10^5	3.62×10^5
	20	5.34×10^5	$5.28 (6.60) \times 10^5$	4.41×10^5	7.57×10^5
	30	3.06×10^5	$2.97 (3.74) \times 10^5$	2.44×10^5	5.47×10^5
	40	6.39×10^4	$5.26 (6.68) \times 10^4$	4.33×10^4	1.72×10^5

(continued)

TABLE I. (Continued)

$E(\infty)^b$	J^c	Sum of States ^a			
		Flexible ^d	Hindered ^e	Harmonic ^f	Free ^g
9.775	0	1.36×10^3	$1.11 (1.43) \times 10^3$	9.72×10^2	1.33×10^3
	10	4.29×10^5	$3.84 (4.71) \times 10^5$	3.32×10^5	4.85×10^5
	20	7.34×10^5	$7.36 (9.15) \times 10^5$	6.04×10^5	9.87×10^5
	30	4.68×10^5	$4.72 (6.15) \times 10^5$	3.59×10^5	7.83×10^5
	40	1.16×10^5	$8.22 (10.6) \times 10^4$	7.62×10^4	2.78×10^5
11.476	0	2.21×10^3	$1.83 (2.28) \times 10^3$	1.58×10^3	2.28×10^3
	10	6.99×10^5	$6.51 (8.19) \times 10^5$	5.52×10^5	8.45×10^5
	20	1.28×10^6	$1.29 (1.63) \times 10^6$	1.04×10^6	1.73×10^6
	30	8.94×10^5	$9.13 (11.4) \times 10^5$	7.02×10^5	1.48×10^6
	40	2.89×10^5	$2.53 (3.27) \times 10^5$	2.00×10^5	6.67×10^5
12.705	0	3.10×10^3	$2.61 (3.16) \times 10^3$	2.20×10^3	3.16×10^3
	10	9.90×10^5	$9.17 (11.4) \times 10^5$	7.72×10^5	1.17×10^6
	20	1.86×10^6	$1.87 (2.36) \times 10^6$	1.52×10^6	2.57×10^6
	30	1.37×10^6	$1.42 (1.83) \times 10^6$	1.09×10^6	2.30×10^6
	40	5.13×10^5	$4.64 (6.19) \times 10^5$	3.60×10^5	1.14×10^6
15.881	0	6.72×10^3	$5.94 (7.39) \times 10^3$	4.72×10^3	7.01×10^3
	10	2.20×10^6	$2.11 (2.63) \times 10^6$	1.70×10^6	2.61×10^6
	20	4.50×10^6	$4.57 (5.78) \times 10^6$	3.62×10^6	6.19×10^6
	30	3.75×10^6	$3.93 (5.10) \times 10^6$	2.97×10^6	6.11×10^6
	40	1.67×10^6	$1.64 (2.18) \times 10^6$	1.22×10^6	3.47×10^6
19.057	0	1.30×10^4	$1.11 (1.40) \times 10^4$	9.38×10^3	1.40×10^4
	10	4.51×10^6	$4.20 (5.32) \times 10^6$	3.43×10^6	5.37×10^6
	20	9.88×10^6	$9.78 (12.8) \times 10^6$	7.76×10^6	1.32×10^7
	30	8.83×10^6	$9.37 (12.2) \times 10^6$	7.04×10^6	1.42×10^7
	40	4.43×10^6	$4.52 (6.04) \times 10^6$	3.39×10^6	8.93×10^6
20.084	0	1.60×10^4	$1.38 (1.75) \times 10^4$	1.15×10^4	1.71×10^4
	10	5.55×10^6	$5.21 (6.61) \times 10^6$	4.23×10^6	6.62×10^6
	20	1.24×10^7	$1.22 (1.59) \times 10^7$	9.72×10^6	1.65×10^7
	30	1.13×10^7	$1.19 (1.58) \times 10^7$	9.07×10^6	1.83×10^7
	40	5.87×10^6	$6.13 (8.13) \times 10^6$	4.55×10^6	1.21×10^7
24.436	0	3.51×10^4	$3.04 (3.92) \times 10^4$	2.51×10^4	3.92×10^4
	10	1.23×10^7	$1.16 (1.52) \times 10^7$	9.47×10^6	1.53×10^7
	20	2.83×10^7	$2.94 (3.91) \times 10^7$	2.30×10^7	3.97×10^7
	30	2.80×10^7	$3.14 (4.13) \times 10^7$	2.37×10^7	4.74×10^7
	40	1.65×10^7	$1.84 (2.49) \times 10^7$	1.38×10^7	3.47×10^7
29.324	0	7.56×10^4	$6.78 (8.93) \times 10^4$	5.43×10^4	8.65×10^4
	10	2.71×10^7	$2.59 (3.46) \times 10^7$	2.09×10^7	3.39×10^7
	20	6.73×10^7	$6.84 (9.07) \times 10^7$	5.33×10^7	9.29×10^7
	30	7.35×10^7	$7.85 (10.5) \times 10^7$	5.96×10^7	1.19×10^8
	40	4.87×10^7	$5.38 (7.40) \times 10^7$	3.90×10^7	9.53×10^7
34.211	0	1.51×10^5	$1.36 (1.81) \times 10^5$	1.09×10^5	1.77×10^5
	10	5.65×10^7	$5.34 (7.09) \times 10^7$	4.24×10^7	7.02×10^7
	20	1.46×10^8	$1.43 (1.92) \times 10^8$	1.12×10^8	1.98×10^8
	30	1.62×10^8	$1.77 (2.39) \times 10^8$	1.34×10^8	2.66×10^8
	40	1.15×10^8	$1.32 (1.84) \times 10^8$	9.62×10^7	2.28×10^8
39.702	0	3.09×10^5	$2.72 (3.66) \times 10^5$	2.18×10^5	3.61×10^5
	10	1.12×10^8	$1.09 (1.46) \times 10^8$	8.58×10^7	1.46×10^8
	20	2.99×10^8	$3.02 (4.09) \times 10^8$	2.35×10^8	4.20×10^8
	30	3.60×10^8	$3.95 (5.43) \times 10^8$	2.97×10^8	5.88×10^8
	40	2.79×10^8	$3.17 (4.48) \times 10^8$	2.32×10^8	5.43×10^8

(continued)

TABLE I. (Continued)

$E(\infty)^b$	J^c	Sum of States ^a			
		Flexible ^d	Hindered ^e	Harmonic ^f	Free ^g
47.642	0	7.46×10^5	$6.77 (9.37) \times 10^5$	5.24×10^5	8.99×10^5
	10	2.71×10^8	$2.70 (3.69) \times 10^8$	2.09×10^8	3.65×10^8
	20	7.56×10^8	$7.80 (10.8) \times 10^8$	5.94×10^8	1.09×10^9
	30	9.80×10^8	$1.07 (1.52) \times 10^9$	8.01×10^8	1.60×10^9
	40	8.29×10^8	$9.48 (13.7) \times 10^8$	6.89×10^8	1.60×10^9
55.582	0	1.61×10^6	$1.47 (2.09) \times 10^6$	1.14×10^6	1.99×10^6
	10	6.05×10^8	$5.93 (8.42) \times 10^8$	4.57×10^8	8.14×10^8
	20	1.73×10^9	$1.76 (2.52) \times 10^9$	1.34×10^9	2.49×10^9
	30	2.34×10^9	$2.54 (3.68) \times 10^9$	1.90×10^9	3.84×10^9
	40	2.07×10^9	$2.41 (3.58) \times 10^9$	1.76×10^9	4.06×10^9
63.522	0	3.23×10^6	$2.93 (4.27) \times 10^6$	2.28×10^6	4.05×10^6
	10	1.23×10^9	$1.19 (1.75) \times 10^9$	9.24×10^8	1.67×10^9
	20	3.69×10^9	$3.62 (5.35) \times 10^9$	2.77×10^9	5.21×10^9
	30	5.31×10^9	$5.44 (8.21) \times 10^9$	4.08×10^9	8.32×10^9
	40	4.94×10^9	$5.41 (8.37) \times 10^9$	4.00×10^9	9.19×10^9

^aThe sum of states reported is the total sum of states for a particular E and J . For the model proposed here it is $N(E, J) = (2J + 1)N_K(E, J)$. If no number is entered for the sum of states, there is no active energy for the model.

^bTotal energy (kcal/mol) of $H + CH_3$ above their zero-point level.

^cIf external rotation is treated classically with $RT/2$ about each axis and the total angular momentum equated to $\sqrt{J(J+1)}\hbar$, it is found that $J = 10, 20, 30$, and 40 corresponds to $T = 566, 2159, 4781$, and 8432 K, respectively.

^dFlexible transition state model; refs. [2,24–26]. The frequencies for the conserved modes were found by reaction path following; ref. [28].

^eThis work with $V(\theta) = V_0 \sin^2 \theta$.

^fThe transitional modes are treated as harmonic oscillators whose frequencies are determined by following the reaction path on the analytic potential energy surface; refs. [18] and [40].

^gThis work with $V(\theta) = 0$.

^hThe rate constants in parentheses were calculated using I_{CH_3} instead of I in eq. (11).

A comparison of the sum of states in Table I shows that, overall, the hindered rotor model is in better agreement with the flexible transition state model than is the harmonic oscillator model. The average percent difference between the flexible and hindered rotor modes is 23, 6, 7, 23, and 37 for $J = 0, 10, 20, 30$, and 40 . For the flexible and harmonic oscillator models the percent differences are 38, 26, 28, 29, and 38 for the same respective values of J . Thus, it is for $J \leq 30$ where the hindered rotor model significantly improves the comparison with the flexible model.

Using I_{CH_3} instead of I in eq. (11) for the calculations (see the discussion above) increases the hindered rotor and free rotor rate constants since the rate constant is proportional to the rotor's moment of inertia. This effect is more important at large E and J than at small E and J . For the latter condition the $H\cdots CH_3$ distance and the transition state is loose so that $I_{H\cdots CH_3}$ is much larger than I_{CH_3} . Thus, I and I_{CH_3} are approximately the same. At larger E and J the transition state tightens and I becomes less than $I_{H\cdots CH_3}$. To illustrate, calculations for the hindered rotor model at $E = 0.4428$ kcal/mol and $J = 0$ give $I_{CH_3}/I = 1.14$ at the transition state. However, for the largest E and J considered here, i.e., 63.522 kcal/mol and 40 , respectively, $I_{CH_3}/I = 1.55$ at the hindered rotor transition state. Rate constants calculated using I_{CH_3} instead of I for the hindered rotor model are given in parentheses in Table I.

The hindered rotor potential is $V(\theta) = V_o \sin^2\theta$ where V_o is given by eq. (12). The degenerate transitional mode harmonic oscillator frequencies for this potential are found from small displacements and are given by eq. (A9). This equation will not give the exact harmonic transitional mode frequencies along the reaction path, since $V(\theta)$ is not an exact fit to the θ -potential of the $\text{CH}_4 \rightleftharpoons \text{H} + \text{CH}_3$ analytic potential energy surface and the Hamiltonian in eq. (3) assumes the θ and ϕ motions are uncoupled from the internal motions of the CH_3 moiety. Nevertheless, it is still of interest to determine how well the transitional mode harmonic frequency from eq. (A9) agrees with the correct value for this frequency determined from reaction path following. The comparison is made in Table II. The difference is large (ca. 40%) for positions along the reaction path with large H---CH₃ distances, but small (less than 10%) for H---CH₃ distances less than 3.4 Å. The frequencies given in parentheses were calculated using I_{CH_3} instead of I in eq. (11). In previous work Aubanel, et al. [29] listed frequencies calculated from eq. (A9) using I_{CH_3} , for H---CH₃ distances corresponding to canonical transition states.

TABLE II. The H---CH₃ transitional mode frequency determined accurately from a reaction path analysis and approximately from the hindered rotor Hamiltonian.

$R(\text{\AA})^a$	Transitional Mode Frequency (cm^{-1})	
	Reaction Path ^b	Hindered Rotor ^c
2.0	827	861 (626) ^d
2.1	749	757 (573)
2.2	670	669 (523)
2.3	600	591 (474)
2.4	534	523 (429)
2.5	470	462 (386)
2.6	414	407 (346)
2.7	360	358 (308)
2.8	313	315 (274)
2.9	272	275 (242)
3.0	232	240 (213)
3.1	200	209 (187)
3.2	171	181 (163)
3.3	144	156 (142)
3.4	122	135 (123)
3.5	102	115 (106)
3.6	86	99 (92)
3.7	71	84 (78)
3.8	61	71 (66)
3.9	49	60 (56)
4.0	40	51 (48)
4.1	33	42 (40)
4.2	27	35 (33)
4.3	22	29 (27)
4.4	18	24 (23)
4.5	16	20 (19)

^aH---CH₃ bond distance at a position along the reaction path.

^bAccurate transitional mode harmonic frequency obtained by reaction path following and calculating the 3N-7 harmonic frequencies orthogonal to the reaction path; refs. [18] and [40].

^cApproximate transitional mode harmonic frequency obtained from the hindered rotor Hamiltonian, i.e., eqs. (3) and (A9).

^dThe frequencies in parentheses were calculated using I_{CH_3} instead of I in eq. (11).

Summary

In this work a model is described for performing variational RRKM calculations. The model combines an analytic treatment of the classical density of states for transitional modes with a numerical quantum density of states for conserved modes. Reaction path properties such as potential energy, external rotation moments of inertia, harmonic frequencies for the $3N-7$ modes orthogonal to the reaction path, and potential energy barriers for internal rotations are incorporated into the model. The specific findings of this study are summarized in the following: (i) In previous work Pacey [34–36] suggested representing the potential energy for rotation of a unimolecular fragment moiety by the function $V(\theta) = V_0 \sin^2\theta$. Here it is shown that, when this potential is incorporated into a two-dimensional Hamiltonian for rotation of the fragment moiety [eq. (3)], an analytic expression is found for the classical density of states of this rotation; (ii) Harmonic frequencies for the $\text{H} \cdots \text{CH}_3$ transitional modes were calculated from an analytic potential energy function for the $\text{CH}_4 \rightleftharpoons \text{H} + \text{CH}_3$ reactive system [21,39,40]. The two-dimensional rotational Hamiltonian, with the Pacey model for $V(\theta)$ fit to the analytic potential function, gives harmonic frequencies for the transitional modes which are in overall good agreement with the exact values found from a reaction path analysis for the analytic function; (iii) Variational transition states and their sums of states for the $\text{CH}_4 \rightleftharpoons \text{H} + \text{CH}_3$ analytic potential energy function have been calculated previously using the flexible transition state theory (FTST) model for the transitional modes and treating the transitional modes as harmonic oscillators [8]. The hindered rotor model developed here gives unimolecular rate constants in better agreement with those of FTST than does a harmonic oscillator model for the transitional modes. For $J = 10$, which corresponds to a rotational temperature of 566 K, the hindered rotor rate constants differ by 6 percent with those of the FTST model, while the harmonic oscillator rate constants differ by 26 percent; and (iv) The hindered rotor model developed here gives rate constants for the $\text{CH}_4 \rightleftharpoons \text{H} + \text{CH}_3$ system which are smaller than those of the FTST model. This is somewhat surprising since the FTST model places the proper constraints on the transition state angular momentum, while the hindered rotor model does not. For the hindered rotor model external rotation is treated as a symmetric (or near-symmetric) top with proper limits placed on the K quantum number. However, the angular momentum of the hindered rotation of the CH_3 moiety is not coupled to the external rotational angular momentum. Properly constraining the angular momentum should reduce the number of energy states (or phase space), which is expected to lead to a smaller rate constant. If I_{CH_3} is used instead of I in eq. (11) for the hindered rotor moment of inertia, the hindered rotor rate constant becomes larger than the FTST rate constant for many E and J . However, there still are numerous E and J for which the hindered rotor rate constants are smaller than the FTST values.

In the same vein, it is somewhat surprising that the rate constants for the FTST model are larger than those for the model which treats the transitional modes as harmonic oscillators. For some reactions, a possible difficulty with the harmonic oscillator treatment is that small frequencies for the transitional modes can lead to a divergent density of states. Apparently, this is not a problem for the $\text{CH}_4 \rightleftharpoons \text{H} + \text{CH}_3$ system.

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Appendix

The classical partition function for the Hamiltonian in eq. (3) and the potential $V(\theta) = V_o \sin^2 \theta$ is [48]

$$\begin{aligned}
 (A1) \quad Q(T) &= \frac{1}{\sigma h^2} \int_0^\pi d\theta \int_0^{2\pi} d\phi \int_{-\infty}^\infty dp_\theta \int_{-\infty}^\infty dp_\phi \\
 &\quad \times \exp[-(\rho_\theta^2/2I + \rho_\phi^2/2I \sin^2 \theta + V_o \sin^2 \theta)/kT] \\
 &= \frac{4\pi^2 I kT}{\sigma h^2} \int_0^\pi \exp(-V_o \sin^2 \theta/kT) \sin \theta d\theta
 \end{aligned}$$

Since the partition function is the Laplace transform of the density of states, the density of states is the following inverted Laplace transform of the partition functions

$$\begin{aligned}
 (A2) \quad \rho(E) &= \mathcal{L}^{-1}[Q(T)] \\
 &= \frac{4\pi^2 I}{\sigma h^2} \mathcal{L}^{-1} \left\{ (kT) \int_0^\pi \exp(-V_o \sin^2 \theta/kT) \sin \theta d\theta \right\}
 \end{aligned}$$

Exchanging the order of the integration and Laplace transform gives

$$\begin{aligned}
 (A3) \quad \rho(E) &= \frac{4\pi^2 I}{\sigma h^2} \int_0^\pi \sin \theta \{ \mathcal{L}^{-1}[kT \exp(-V_o \sin^2 \theta/kT)] \} d\theta \\
 &= \frac{4\pi^2 I}{\sigma h^2} \int_0^\pi \sin \theta \Delta(E - V_o \sin^2 \theta) d\theta
 \end{aligned}$$

where

$$\begin{aligned}
 (A4) \quad \Delta(E - V_o \sin^2 \theta) &= 1, \quad E \geq V_o \sin^2 \theta \\
 &= 0, \quad E < V_o \sin^2 \theta
 \end{aligned}$$

There are the following two cases for eq. (A3).

1. $E \geq V_o$. For this situation, $\Delta(E - V_o \sin^2 \theta)$ equals unity so that the density is

$$(A5) \quad \rho(E) = \frac{8\pi^2 I}{\sigma h^2}$$

which is the density of states for a two-dimensional free rotor.

2. $E < V_o$. To solve eq. (3) for this case a limiting angle θ_a is found when $V_o \sin^2 \theta$ equals E , i.e.,

$$(18) \quad E - V_o \sin^2 \theta_a = 0$$

Thus, $\Delta(E - V_o \sin^2 \theta)$ is unity and nonzero for θ in the ranges $0 \leq \theta \leq \theta_a$ and $\theta_a \leq \theta \leq \pi$. The density of states is then

$$(A7) \quad \rho(E) = [8\pi^2 I / \sigma h^2] [1 - (1 - E/V_o)^{1/2}]$$

For $E \ll V_o$, eq. (A7) becomes

$$(A8) \quad \rho(E) = 4\pi^2 I E / (\sigma h^2 V_o)$$

which is the harmonic oscillator density of states for a two-dimensional rotor with a degenerate frequency of [29,36]

$$(A9) \quad \nu = (\sigma V_o / I)^{1/2} / 2\pi.$$

Bibliography

- [1] W.L. Hase, *Acc. Chem. Res.*, **16**, 258 (1983).
- [2] D.M. Wardlaw and R.A. Marcus, *Adv. Chem. Phys.*, **70**, 231 (1988).
- [3] W.L. Hase and D.M. Wardlaw, in *Bimolecular Collisions*, M.N.R. Ashfold and J.E. Baggot, Eds., Royal Society of Chemistry, London, 1989, p. 171.
- [4] M. Quack and J. Troe, *Ber. Bunsenges, Phys. Chem.*, **78**, 240 (1974).
- [5] W.H. Miller, *J. Am. Chem. Soc.*, **101**, 6810 (1979).
- [6] J. Troe, *J. Chem. Phys.*, **79**, 6017 (1983).
- [7] L. Zhu and W.L. Hase, *Chem. Phys. Lett.*, **175**, 117 (1990).
- [8] E.E. Aubanel, D.M. Wardlaw, L. Zhu, and W.L. Hase, *Int. Rev. Phys. Chem.*, **10**, 249 (1991).
- [9] L. Zhu, W. Chen, W.L. Hase, and E.W. Kaiser, *J. Phys. Chem.*, **97**, 311 (1993).
- [10] R.G. Gilbert and S.C. Smith, *Theory of Unimolecular and Recombination Reactions*, Blackwell Scientific, Oxford, 1990.
- [11] S.C. Smith, *J. Chem. Phys.*, **95**, 3404 (1991).
- [12] S.C. Smith, *J. Chem. Phys.*, **97**, 2406 (1992).
- [13] S.C. Smith, *J. Phys. Chem.*, submitted.
- [14] In addition to the two variational RRKM models considered here, two additional microcanonical models have been proposed which use a limited amount of potential energy surface information. A scheme which utilizes some potential energy surface information has been proposed for interpolating between a unimolecular reactant vibrational partition function and a fragment rotational partition function; W. Forst, *J. Phys. Chem.*, **95**, 3612 (1991); W. Forst and F. Caralp, *J. Chem. Soc., Faraday Trans.*, **87**, 2307 (1991); W. Forst and F. Caralp, *J. Phys. Chem.*, **96**, 6291 (1992). Some potential energy surface information may be used in constructing adiabatic curves for the statistical adiabatic channel model; M. Quack and J. Troe, *Ber. Bunsenges. Phys. Chem.*, **78**, 240 (1974); **81**, 329 (1977).
- [15] K. Fukui, S. Kato, and H. Fujimoto, *J. Am. Chem. Soc.*, **97**, 1 (1975); S. Kato, H. Kato, and K. Fukui, *ibid.*, **99**, 684 (1977); H.F. Schaefer, *Chem. Br.*, **11**, 227 (1975); K. Ishida, K. Morokuma, and A. Komornicki, *J. Chem. Phys.*, **66**, 2153 (1977).
- [16] W.H. Miller, N.C. Handy, and J.E. Adams, *J. Chem. Phys.*, **72**, 99 (1980).
- [17] S. Kato and K. Morokuma, *J. Chem. Phys.*, **73**, 3900 (1980); K. Morokuma and S. Kato in *Potential Energy Surfaces and Dynamics Calculations*, D.G. Truhlar, Ed., Plenum, New York, 1981, p. 243.
- [18] W.L. Hase and R.J. Duchovic, *J. Chem. Phys.*, **83**, 3448 (1985).
- [19] S.L. Mondro, S.R. Vande Linde, and W.L. Hase, *J. Chem. Phys.*, **84**, 3783 (1986).
- [20] S.R. Vande Linde, S.L. Mondro, and W.L. Hase, *J. Chem. Phys.*, **86**, 1348 (1987).
- [21] W.L. Hase, S.L. Mondro, R.J. Duchovic, and D.M. Hirst, *J. Am. Chem. Soc.*, **109**, 2916 (1987).
- [22] W.L. Hase, M.-C. Richou, and S.L. Mondro, *J. Phys. Chem.*, **93**, 539 (1989).
- [23] Y.N. Lin and B.S. Rabinovitch, *J. Phys. Chem.*, **74**, 3151 (1970).
- [24] D.M. Wardlaw and R.A. Marcus, *Chem. Phys. Lett.*, **110**, 230 (1984).
- [25] D.M. Wardlaw and R.A. Marcus, *J. Chem. Phys.*, **83**, 3462 (1985).
- [26] D.M. Wardlaw and R.A. Marcus, *J. Phys. Chem.*, **90**, 5383 (1986).
- [27] W.J. Chesnavich and M.T. Bowers, in *Gas Phase Ion Chemistry*, M.T. Bowers, Ed., Academic Press, New York, 1979, p. 119, Vol. I.
- [28] E.E. Aubanel and D.M. Wardlaw, *J. Phys. Chem.*, **93**, 3117 (1989).
- [29] E.E. Aubanel, S.H. Robertson, and D.M. Wardlaw, *J. Chem. Soc., Faraday Trans.*, **87**, 2291 (1991).
- [30] S.J. Klippenstein, *Chem. Phys. Lett.*, **170**, 71 (1990).
- [31] S.J. Klippenstein, *J. Chem. Phys.*, **94**, 6469 (1991).
- [32] S.J. Klippenstein, *J. Chem. Phys.*, **96**, 367 (1992).
- [33] M.J.T. Jordan, S.C. Smith, and R.G. Gilbert, *J. Phys. Chem.*, **95**, 8685 (1991).
- [34] P.D. Pacey, *J. Chem. Phys.*, **77**, 3540 (1982).
- [35] J.F. LeBlanc and P.D. Pacey, *J. Chem. Phys.*, **83**, 4511 (1985).
- [36] S.C. King, J.F. LeBlanc, and P.D. Pacey, *Chem. Phys.*, **123**, 329 (1988).
- [37] If the transitional modes consist of multiple rotors, as the two 2-dimensional rotors and free rotor for C_2H_6 dissociation, analytic expressions for individual rotor densities of states are convoluted to obtain the total transitional mode density of states. Convolutions are defined in W. Forst, *Theory of Unimolecular Reactions*, Academic Press, New York, 1973.

- [38] T. Beyer and D. F. Swinehart, *Commun. Assoc. Compt. Machin.*, **16**, 372 (1973); S. E. Stein and B. S. Rabionovitch, *J. Chem. Phys.*, **58**, 2438 (1973).
- [39] R. J. Duchovic, W. L. Hase, and H. B. Schlegel, *J. Phys. Chem.*, **88**, 1339 (1984).
- [40] X. Hu and W. L. Hase, *J. Chem. Phys.*, **95**, 8073 (1991).
- [41] C. J. Cobos and J. Troe, *Z. Phys. Chem. Neue Folge*, **167**, 129 (1990).
- [42] M. Brouard, M. T. Macpherson, and M. J. Pilling, *J. Phys. Chem.*, **93**, 4049 (1989).
- [43] P. H. Stewart, G. P. Smith, and D. M. Golden, *Int. J. Chem. Kinet.*, **21**, 923 (1989).
- [44] The analytic potential energy surface for the methane system is incorporated in the general chemical dynamics computer program VENUS. Harmonic frequencies and moments of inertia along the path were obtained by using the reaction path following and eigenvalue/eigenvector options in VENUS.
- [45] K. S. Pitzer and W. D. Gwinn, *J. Chem. Phys.*, **10**, 428 (1942).
- [46] S. J. Klippenstein and R. A. Marcus, *J. Chem. Phys.*, **87**, 3410 (1987).
- [47] The $N^{\ddagger}(E, J, K)$ in eq. (10) were found numerically from eq. (8) using 8-point gaussian integration.
- [48] The approach used here, for finding the transitional mode density of states for the Pacey function $V(\theta) = V_0 \sin^2 \theta$, has also been used in canonical variational transition state theory calculations; S. H. Robertson and D. M. Wardlaw, private communication.

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